

Project Name : Gold Trading Automation Platform (GTAP)

Document Name : Business Requirement Specification (BRS)

Version : 1.0

Prepared By : Vinit Kushwaha

Date : 05 July 2026

Status : Draft

Business Requirement Specification (BRS)

Version 1.

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Chapter 1 - Cover Page

Business Requirement Specification (BRS)

Project Name

Gold Trading Automation Platform (GTAP)

Document Type

Business Requirement Specification (BRS)

Version

Version 1.0

Project Type

Automated Gold Trading Platform

Client

Confidential

Prepared By

Vinit Kushwaha

Technology Stack

- Node.js
- Express.js

- MongoDB
- Python
- MetaTrader 5
- Socket.IO
- Telegram Bot API
- HTML
- CSS
- JavaScript

Date

05 July 2026



Chapter 2 – Document Control

Document Information

Field	Value
Document Name	Business Requirement Specification
Version	1.0
Status	Draft
Prepared By	Vinit Kushwaha
Project	Gold Trading Automation Platform

Version History

Version	Date	Changes
1.0	05 July 2026	Initial Draft

Approval

Role	Name	Signature
Client	Pending	Pending
Developer	Vinit Kushwaha	Pending



Chapter 3 – Project Overview

3.1 Introduction

The Gold Trading Automation Platform (GTAP) is a web-based automated trading system designed to execute Gold (XAUUSD) trades automatically on MetaTrader 5 based on a predefined scalping strategy.

The system minimizes manual intervention by continuously monitoring market conditions and executing Buy/Sell orders according to the configured strategy.

3.2 Problem Statement

Currently, the trader manually monitors charts and executes trades based on market conditions. This approach has several limitations:

- Human delay in execution
- Emotional trading decisions
- Missed trading opportunities
- Lack of centralized monitoring
- Difficulty in maintaining consistent execution

The project aims to eliminate these issues through complete automation.

3.3 Project Objective

The primary objective is to develop a secure, reliable, and scalable trading platform capable of:

- Executing Buy/Sell orders automatically
- Monitoring market conditions in real time
- Managing Stop Loss and Take Profit
- Providing a real-time dashboard

- Sending Telegram notifications
 - Supporting future scalability
-

3.4 Project Vision

To build a professional automated trading platform that delivers consistent execution, minimizes manual effort, and serves as a scalable foundation for future multi-user trading services.

3.5 Project Mission

To provide traders with an intelligent automation platform capable of executing predefined strategies with high reliability, low latency, and comprehensive trade monitoring.



Chapter 4 – Business Objectives

The business objectives of this project are:

Primary Objectives

- Fully automate Buy orders.
 - Fully automate Sell orders.
 - Eliminate manual trading decisions.
 - Reduce execution delay.
 - Maintain low drawdown through disciplined risk management.
 - Provide real-time portfolio monitoring.
 - Send instant Telegram notifications.
 - Support both Demo and Real MT5 accounts.
-

Secondary Objectives

- Design a scalable architecture.
- Enable future support for multiple users.
- Enable future support for multiple trading accounts.
- Enable future support for multiple strategies.
- Provide professional reporting and analytics.
- Prepare the platform for future commercial deployment.

Chapter 5 – Project Scope

5.1 Project Scope

Purpose

The purpose of the **Gold Trading Automation Platform (GTAP)** is to automate the execution of Gold (XAUUSD) trades on MetaTrader 5 using a predefined scalping strategy. The platform will minimize manual intervention while providing real-time monitoring, risk management, and reporting capabilities.

5.2 In Scope (MVP)

The first version (MVP) of the system will include the following features:

Trading

- Automated Buy Execution
 - Automated Sell Execution
 - XAUUSD (Gold) Trading Only
 - MetaTrader 5 Integration
 - XM Broker Support
 - Scalping Strategy
 - EMA (11)
 - RSI (14)
 - 5-Candle Fractal Detection
 - Stop Loss
 - Take Profit (1:2 Risk Reward)
 - Trailing Stop Loss
 - Maximum 5 Active Trades
 - Fixed Lot Size (0.2)
-

Dashboard

The Dashboard will provide:

- Live Account Balance
 - Equity
 - Margin
 - Free Margin
 - Today's Profit/Loss
 - Weekly Profit
 - Monthly Profit
 - Active Trades
 - Closed Trades
 - Trade History
 - Win Rate
 - Drawdown
 - Strategy Status
 - Bot Status
-

Notifications

- Telegram Notifications
 - Trade Open Alert
 - Trade Close Alert
 - Error Notifications
 - VPS Restart Notification
-

Risk Management

- Low Drawdown
 - Maximum 5 Open Trades
 - Automatic Stop Loss
 - Automatic Take Profit
 - Trailing Stop Loss
 - Retry if Connection Lost
-

Security

- User Login
 - JWT Authentication
 - Secure API Access
-

5.3 Out of Scope (Version 1)

The following features are **not included** in the first release:

- Mobile Application
- Copy Trading
- Payment Gateway
- Multiple Brokers
- AI-Based Strategy
- Voice Commands
- Social Trading

These features may be considered in future versions.

5.4 Future Expansion

The system architecture should support future enhancements without requiring a complete redesign.

Future enhancements may include:

- Multiple Trading Accounts
 - Multiple Users
 - Admin Panel
 - Subscription System
 - Additional Trading Strategies
 - AI-Based Analytics
 - Mobile Application
 - Performance Reports
 - Broker Management
-



Chapter 6 – Stakeholders

6.1 Primary Stakeholders

Client

Responsible for:

- Providing Trading Strategy
- Validating Business Requirements

- Approving Features
 - Testing Final Product
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Developer

Responsible for:

- Software Design
 - Development
 - Testing
 - Deployment
 - Maintenance
-

Trader

Responsible for:

- Managing Trading Account
 - Monitoring Dashboard
 - Reviewing Reports
 - Configuring Trading Parameters
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6.2 Future Stakeholders

- System Administrator
- Support Team
- Additional Traders
- Business Owner



Chapter 7 – Functional Requirements

यह Chapter बताता है कि **System क्या-क्या करेगा।**

7.1 User Authentication

- Secure Login
- JWT Authentication
- Session Management
- Secure Logout

7.2 Trading Engine

The system shall:

- Connect to MetaTrader 5.
 - Monitor XAUUSD (1 Minute Chart).
 - Calculate EMA (11).
 - Calculate RSI (14).
 - Detect 5-Candle Fractals.
 - Execute Buy Orders automatically.
 - Execute Sell Orders automatically.
 - Apply Stop Loss.
 - Apply Take Profit (1:2 RR).
 - Apply Trailing Stop Loss.
 - Retry connection automatically if MT5 disconnects.
-

7.3 Dashboard

The dashboard shall display:

- Account Balance
 - Equity
 - Margin
 - Free Margin
 - Drawdown
 - Active Trades
 - Closed Trades
 - Daily Profit
 - Weekly Profit
 - Monthly Profit
 - Win Rate
 - Bot Status
 - MT5 Connection Status
-

7.4 Notifications

The system shall send Telegram notifications for:

- Bot Started
- Bot Stopped
- Trade Opened

- Trade Closed
 - Stop Loss Hit
 - Take Profit Hit
 - Connection Lost
 - Connection Restored
 - Error Messages
-

7.5 Reports

The system shall provide:

- Daily Report
 - Weekly Report
 - Monthly Report
 - Trade History
 - Performance Report
-

Chapter 8 – Non-Functional Requirements

Performance

- Real-time trade execution
 - Low execution delay
 - Fast dashboard updates
-

Availability

- 24×7 VPS Support
 - Automatic Recovery
 - Auto Reconnection
-

Security

- JWT Authentication

- Password Encryption
 - Secure API Communication
-

Scalability

The system should support:

- Multiple Users
 - Multiple Accounts
 - Multiple Strategies
 - Future Broker Integrations
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Reliability

- Error Logging
 - Retry Mechanism
 - Stable Trading Execution
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Chapter 9 – Business Rules

Trading Rules

- Broker: XM
 - Platform: MetaTrader 5
 - Market: XAUUSD
 - Timeframe: 1 Minute
 - Strategy: Scalping
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Indicators

- EMA = 11
 - RSI = 14
 - 5-Candle Fractal
-

Entry Rules

Buy

- Swing Low detected
- EMA Break
- Confirmation within 3 candles
- Auto Buy

Sell

- Swing High detected
 - EMA Break
 - Confirmation within 3 candles
 - Auto Sell
-

Risk Rules

- Fixed Lot Size = 0.2
 - Maximum Open Trades = 5
 - Stop Loss Buffer = 0.02
 - Risk Reward = 1:2
 - Trailing Stop Loss Enabled
 - Low Drawdown Priority
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Operational Rules

- Weekend Trading = Disabled
 - News Trading = Enabled
 - Demo & Real Accounts Supported
-

Chapter 10 – Assumptions & Risks

Assumptions

- MT5 is installed and configured.
 - VPS is available 24×7.
 - Internet connection is stable.
 - User provides valid MT5 credentials.
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Risks

- Broker downtime
- Internet interruption
- VPS failure
- Market volatility
- MT5 disconnection

Mitigation:

- Auto Retry
 - Telegram Alerts
 - Error Logs
-



Chapter 11 – Future Scope

Future versions may include:

- Multiple Trading Accounts
 - Multiple Users
 - Admin Panel
 - Subscription Plans
 - Payment Gateway
 - AI Strategy
 - Copy Trading
 - Mobile Application
 - Email Notifications
 - Performance Analytics
 - Multi-Broker Support
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Chapter 12 – Approval

Client Approval

Item

Status

Requirements Approved	Pending
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Business Rules Approved	Pending
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Dashboard Approved	Pending
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Developer Approval

Item	Status
BRS Reviewed	Pending
Ready for SRS	Pending